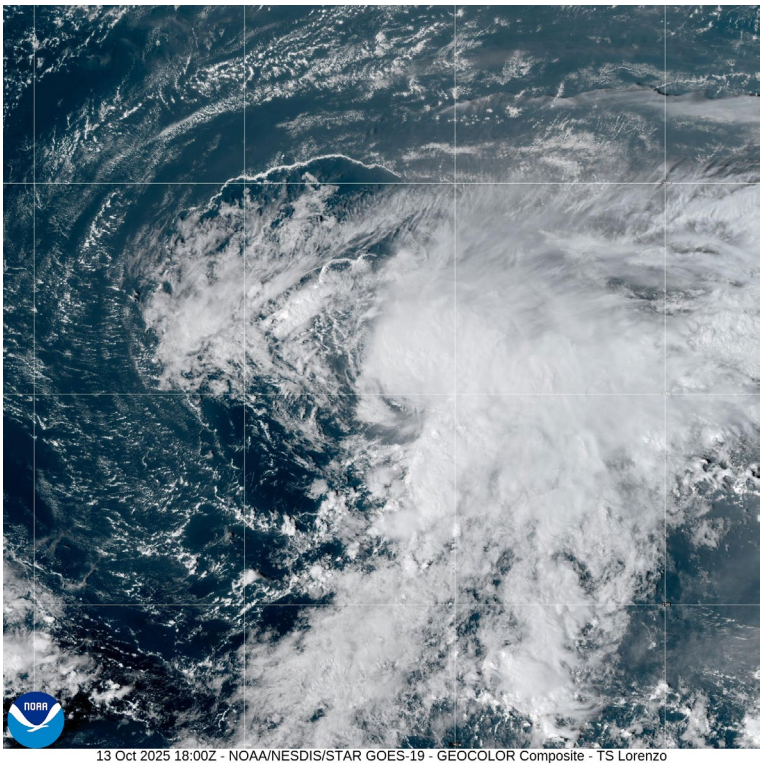
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

TROPICAL STORM LORENZO

(AL122025)

13–15 October 2025

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National Hurricane Center
20 November 2025



GOES-19 GEOCOLOR IMAGE OF LORENZO AT 1800 UTC 13 OCTOBER 2025. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Lorenzo was a tropical storm in the eastern and central tropical Atlantic that stayed far away from land.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Lorenzo.

Tropical Storm Lorenzo

13–15 OCTOBER 2025

BEST TRACK

The "best track[2]" positions and intensities for Tropical Storm Lorenzo are listed in Table 1. The best track chart of Lorenzo's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no surface reports of tropical-storm-force winds from Lorenzo.

Origin

Lorenzo formed from a low-latitude tropical wave that moved westward from the coast of Africa on 9 October. The system first showed signs of convective organization on 11 October, and the next day it developed an area of 35-kt winds to the northeast of a vorticity center embedded within a sharp trough. At about 0600 UTC 13 October, a well-defined circulation developed accompanied by organized convection, and it is estimated that Tropical Storm Lorenzo formed at that time about 900 n mi west of the Cabo Verde Islands.

Peak Intensity and Minimum Pressure

Lorenzo's peak intensity of 50 kt is supported by multiple ASCAT overpasses showing maximum vectors of around 45 kt near 0000 UTC 14 October, along with the likelihood that the maximum winds were undersampled.

The estimated minimum central pressure of 1000 mb is based on the Knaff-Zehr-Courtney wind-pressure relationship for a storm of Lorenzo's intensity and size.

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[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Tropical Analysis and Forecast Branch (TAFB) and the Satellite Analysis Branch (SAB), objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Lorenzo.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Lorenzo.

FORECAST AND WARNING VERIFICATION

Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. Temporally, the development of Lorenzo was not well anticipated, with the system not being introduced into the TWO until 48 h before genesis and the system not being given a high probability of development in the 7-day outlook until the last 6 h before genesis occurred. Figure 4 shows composites of 7-day TWO genesis areas for each category prior to the formation of Lorenzo. While the number of genesis forecasts is low, the genesis location was consistently inside the forecast areas.

A verification of NHC official track forecasts for Lorenzo is given in Table 3a. Official track forecast errors were greater than the mean official errors for the previous 5-yr period at the 12- and 24-h forecast times, but were less than the 5-yr mean errors at the 36- and 48-h forecast times. It should be noted that Lorenzo was only a tropical cyclone for 54 h, so the number of forecasts is small. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. Many of the track guidance models had lower average errors than the official forecast, but again the number of forecasts was small and results are not very meaningful.

A verification of NHC official intensity forecasts for Lorenzo is given in Table 4a. Official intensity forecast errors were greater than the mean official errors for the previous 5-yr period. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. Many of the guidance models had lower average errors than the official forecast at the 24-, 36-, and 48-h forecast times. Examination of the individual forecasts (not shown) indicates that the official intensity forecasts were too low on 13 October and then too high on 14–15 October since Lorenzo weakened faster than originally forecast.

There were no coastal watches or warnings issued for Lorenzo.

ACKNOWLEDGEMENTS

Philippe Papin created the Graphical Tropical Weather Outlook verification graphic.

Table 1. Best track for Tropical Storm Lorenzo, 13–15 October 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
13 / 0600	13.8	39.7	1005	40	tropical storm
13 / 1200	14.4	40.6	1003	45	"
13 / 1800	15.0	41.5	1002	45	"
14 / 0000	15.8	42.4	1000	50	"
14 / 0600	16.9	43.6	1000	50	"
14 / 1200	17.6	44.7	1004	40	"
14 / 1800	18.3	45.3	1005	35	"
15 / 0000	19.2	45.5	1005	35	"
15 / 0600	20.1	45.3	1004	35	"
15 / 1200	21.1	44.5	1004	35	"
15 / 1800					dissipated
14 / 0000	15.8	42.4	1000	50	minimum pressure and maximum winds

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	42	48
Medium (40%-60%)	24	36
High (>60%)	N/A	6

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Lorenzo, 13–15 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	29.3	50.1	**39.3**	**31.3**				
OCD5	60.8	87.4	112.4	167.9				
Forecasts	8	6	4	2				
OFCL (2020-24)	23.0	34.3	45.8	58.7	73.5	89.8	128.7	185.4
OCD5 (2020-24)	45.1	95.7	150.9	203.1	252.7	295.4	366.2	426.6

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Lorenzo, 13–15 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	31.9	51.4	34.2	28.7				
OCD5	62.7	93.3	125.1	183.1				
GFSI	**30.9**	**45.0**	39.3	43.1				
HWFI	39.2	**47.0**	36.1	86.4				
HMNI	45.6	55.8	**27.7**	87.4				
HFAI	33.0	**47.7**	**25.8**	49.2				
HFBI	40.4	**41.7**	**25.5**	41.2				
EMXI	42.3	63.0	52.1	33.6				
CMCI	36.9	**37.0**	46.7	64.1				
TVCA	32.2	**41.0**	**22.5**	50.4				
TVCX	31.9	**39.4**	**22.6**	50.4				
GFEX	**31.1**	**41.6**	**22.4**	39.6				
TVDG	**30.9**	**38.7**	**18.1**	45.2				
HCCA	**31.2**	**41.9**	**28.3**	33.3				
FSSE	32.3	**44.1**	**26.9**	39.2				
AEMI	34.4	**43.3**	34.8	67.2				
TABS	40.0	56.8	60.0	61.6				
TABM	42.7	**50.0**	41.1	137.4				
TABD	46.5	53.4	37.4	123.8				
Forecasts	7	5	3	1				

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Lorenzo, 13–15 October 2025. Mean errors for the previous 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.6	10.0	12.5	15.0				
OCD5	5.6	13.0	21.5	22.5				
Forecasts	8	6	4	2				
OFCL (2020-24)	5.1	7.3	8.6	10.0	10.5	10.9	12.4	13.6
OCD5 (2020-24)	6.8	10.6	13.8	16.5	17.9	19.2	21.4	19.9

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Lorenzo, 13–15 October 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	5.7	10.0	13.3	15.0				
OCD5	6.3	15.4	22.7	24.0				
HWFI	6.4	11.4	14.7	**7.0**				
HMNI	8.7	**8.8**	15.3	**5.0**				
HFAI	5.9	**3.0**	**3.7**	**1.0**				
HFBI	6.9	**3.8**	**2.0**	**3.0**				
DSHP	7.4	13.2	14.7	**14.0**				
LGEM	8.1	12.6	**11.7**	**9.0**				
ICON	7.7	11.6	14.7	**9.0**				
IVCN	7.4	**8.2**	**10.7**	**6.0**				
IVDR	7.3	**7.2**	**9.0**	**5.0**				
GFSI	6.9	**3.8**	**3.7**	**4.0**				
EMXI	7.4	**5.6**	**2.3**	**3.0**				
HCCA	6.9	**8.2**	**8.7**	**6.0**				
FSSE	5.9	**7.4**	**10.3**	**8.0**				
Forecasts	7	5	3	1				

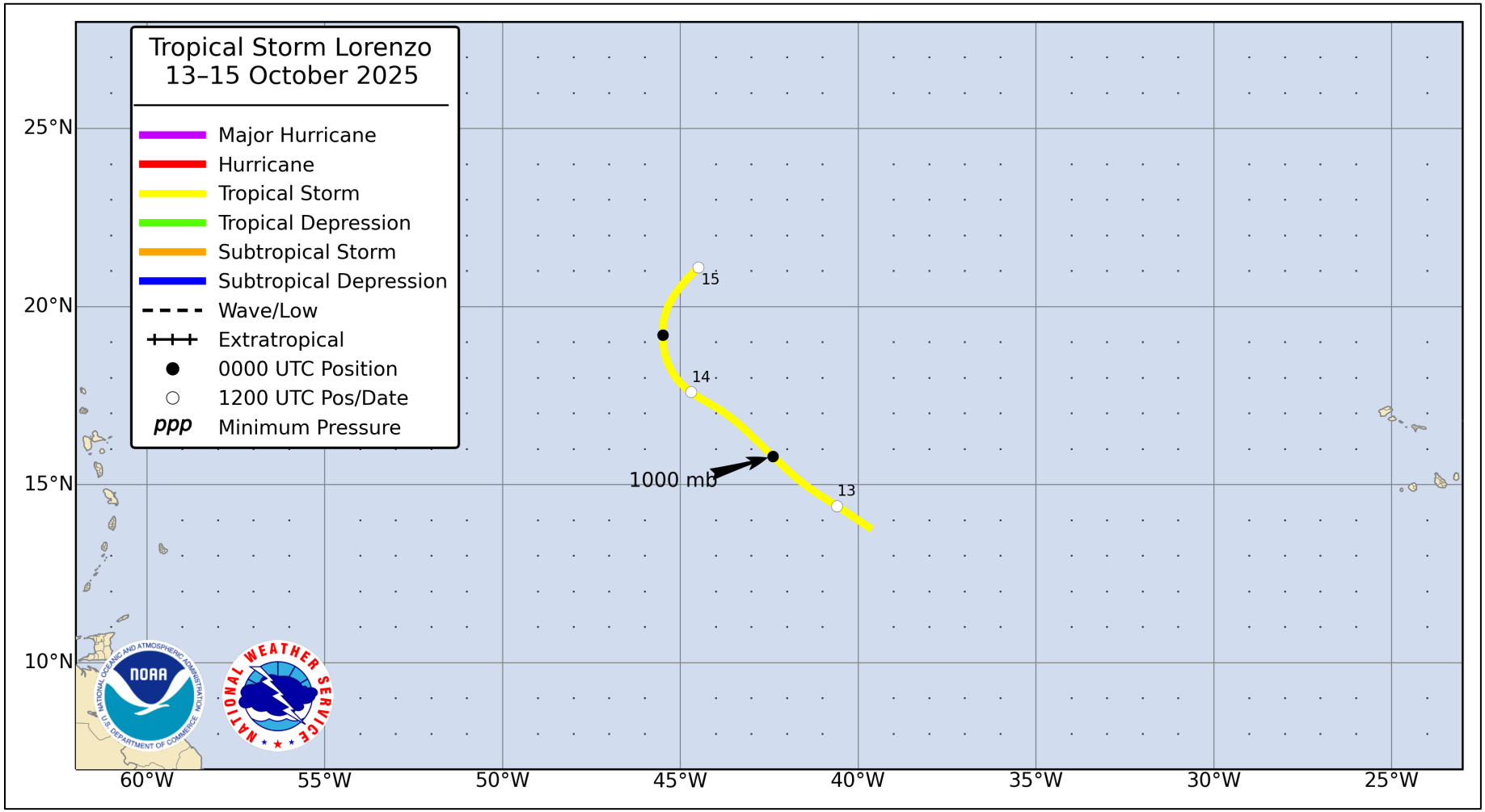


Figure 1. Best track positions for Tropical Storm Lorenzo, 13–15 October 2025.

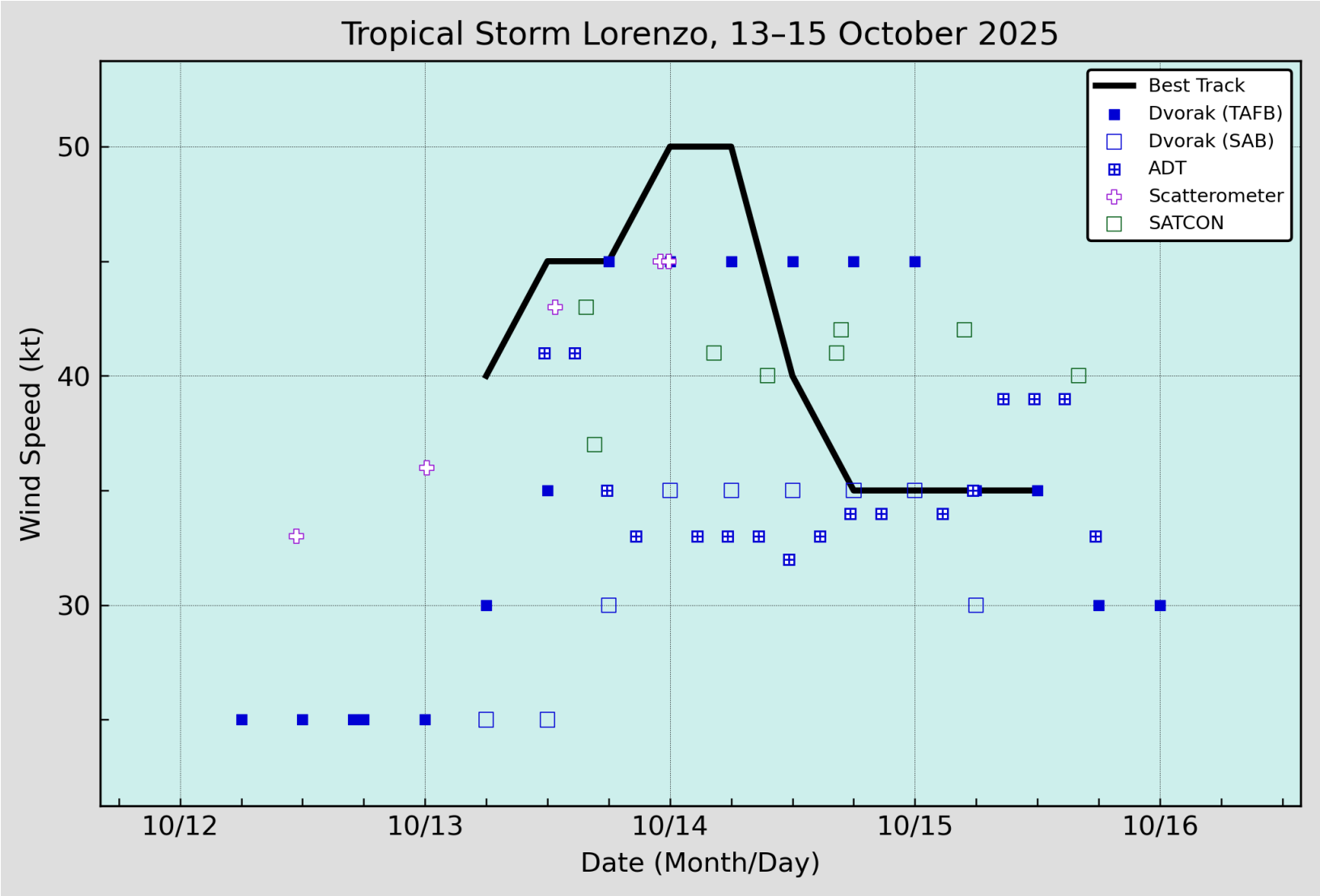


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Lorenzo, 13–15 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

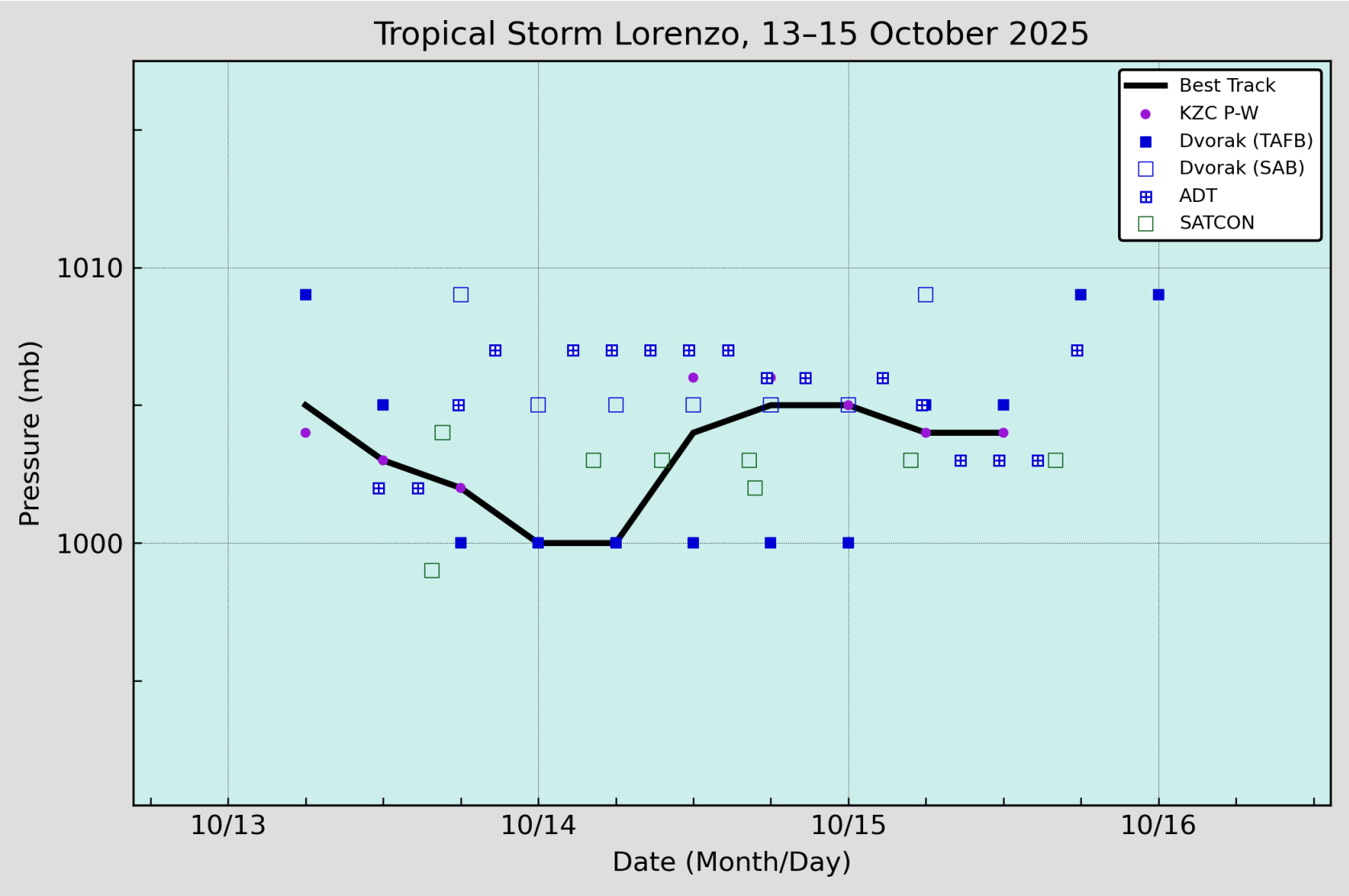


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Lorenzo, 13–15 October 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

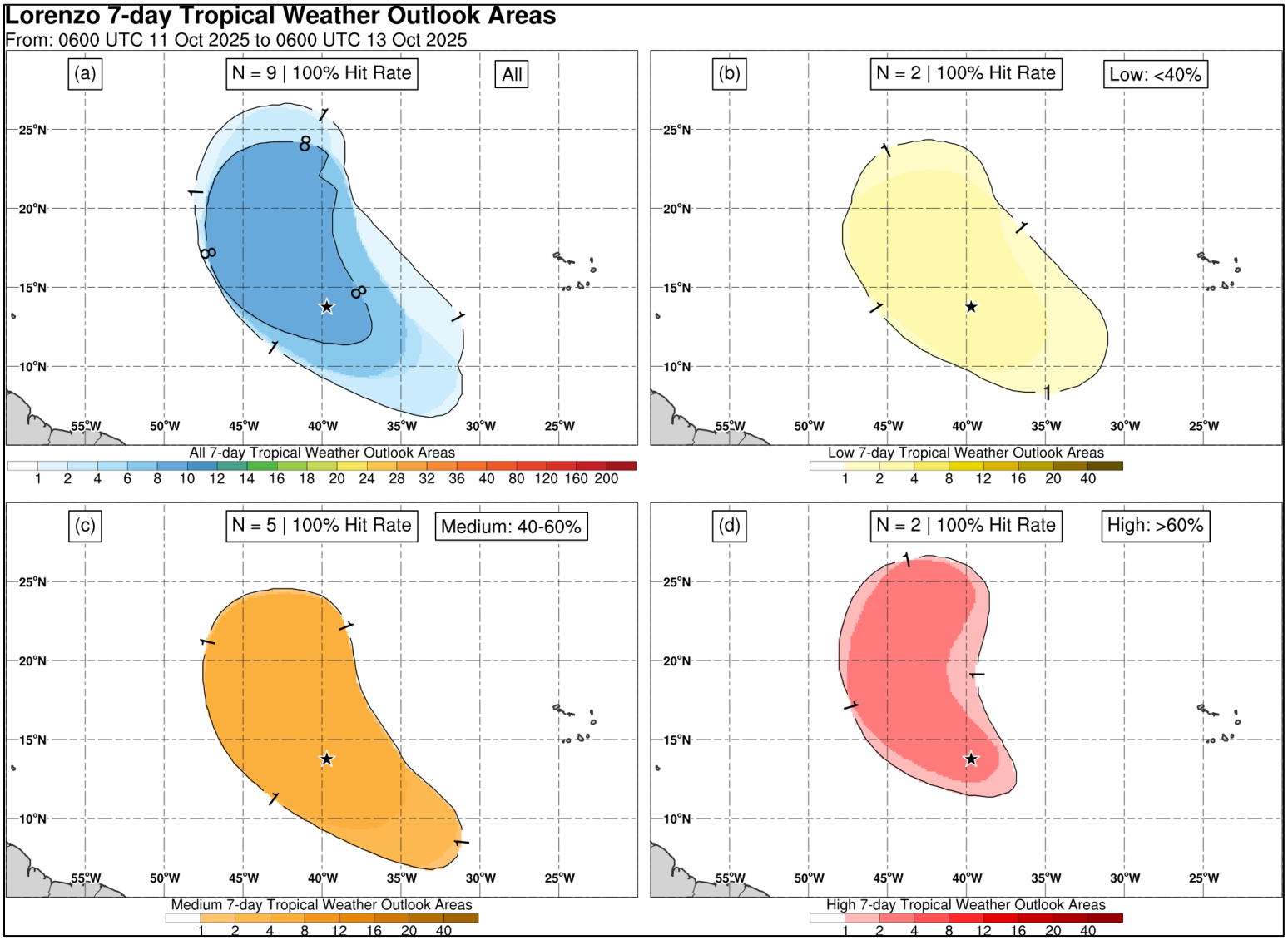


Figure 4. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Lorenzo for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.